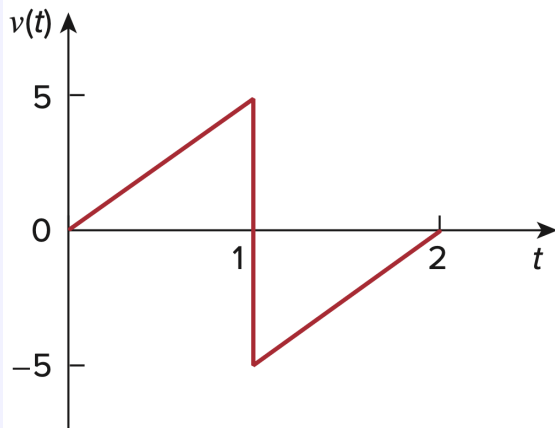


Exercise Sheet #5

Fields 2

Exercise 1: Time-Dependent Inductor Voltage and Current

If the voltage waveform below is applied to a 10-mH inductor, find the inductor current $i(t)$ for $0 < t < 2$ seconds. Assume $i(0) = 0$.



Exercise 2: Coil

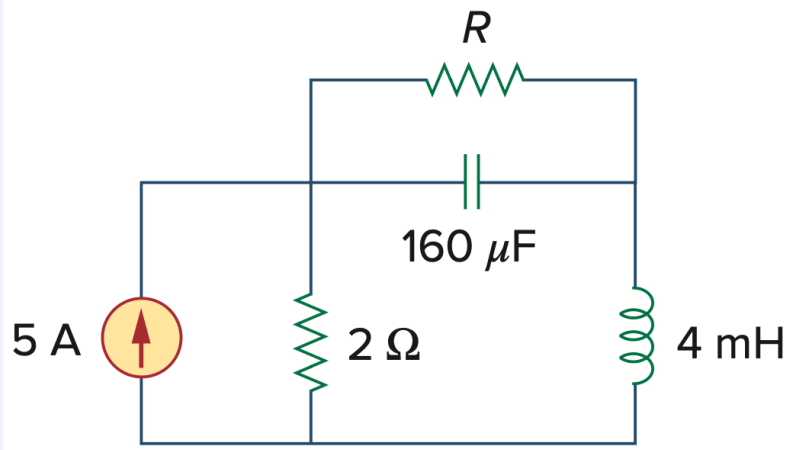
A cylindrical coil with a length of 2.7 m, a radius of 0.85 cm, and 600 turns carries a current of 2.5 A. Calculate B inside the coil, far from its ends.

Exercise 3: Energy of Inductors

The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

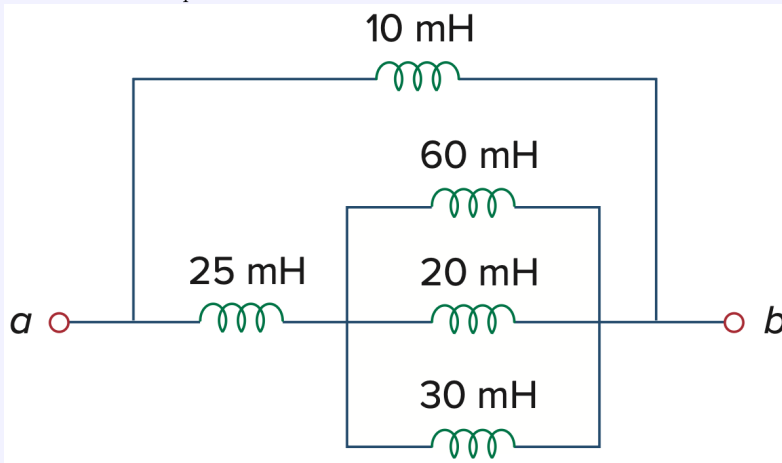
Exercise 4: Inductor and Capacitor under DC

For the circuit below, calculate the value of R that will make the energy stored in the capacitor the same as that stored in the inductor under DC conditions.



Exercise 5: Series and Parallel Inductors

Determine L_{eq} at terminals a-b of the circuit below.

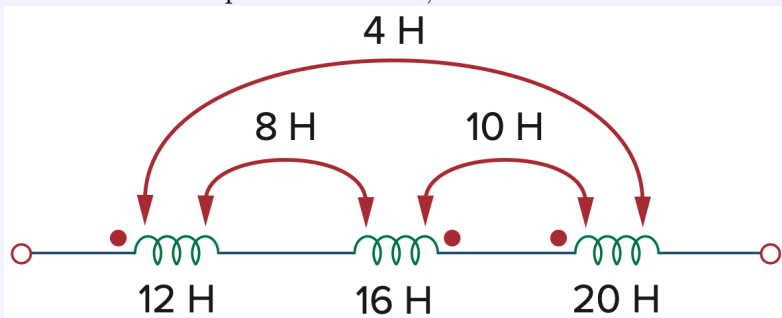
**Exercise 6: Ideal Transformer**

A 480-V to 2400-V step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

- the turns ratio
- the primary current
- the secondary current

Exercise 7: Mutual Inductance

For the three coupled coils below, calculate the total inductance.



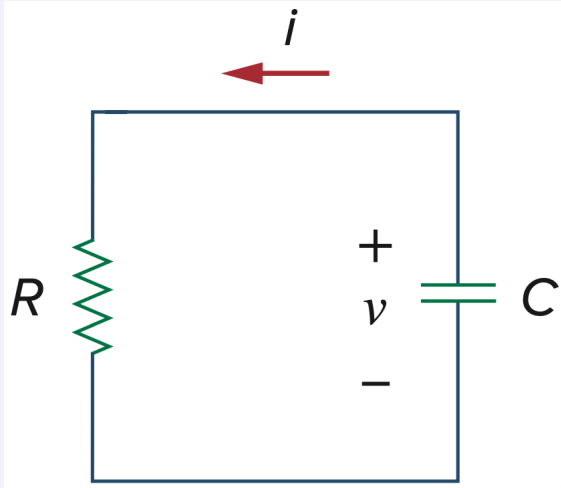
Background: The polarity of the mutual voltage is defined by the orientation or particular way in which the two coils are physically wound. It can be determined by applying Lenz's law in conjunction with the right-hand rule. In circuit diagrams, the polarity is often shown by the dot convention: A dot is placed in the circuit at one end of each of the two magnetically coupled coils to indicate the direction of the magnetic flux if current enters that dotted terminal of the coil. The convention then reads: If a current enters the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is positive at the dotted terminal of the second coil.

Exercise 8: RC-Discharge

In the circuit shown below, for $t > 0$

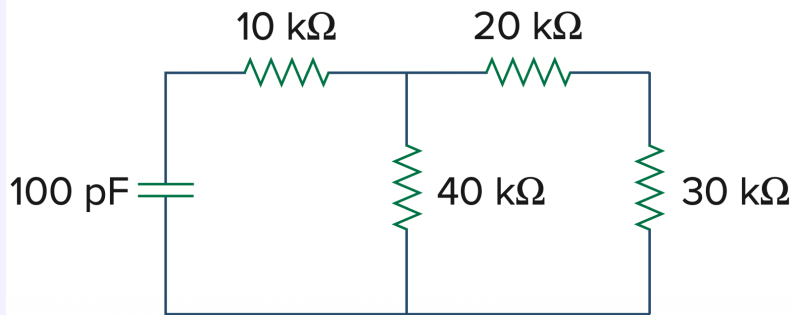
$$v(t) = 56e^{-200t} \text{ V}, \quad i(t) = 8e^{-200t} \text{ mA}.$$

- Find the values of R and C .
- Calculate the time constant τ .
- Determine the time required for the voltage to decay to half its initial value at $t = 0$.



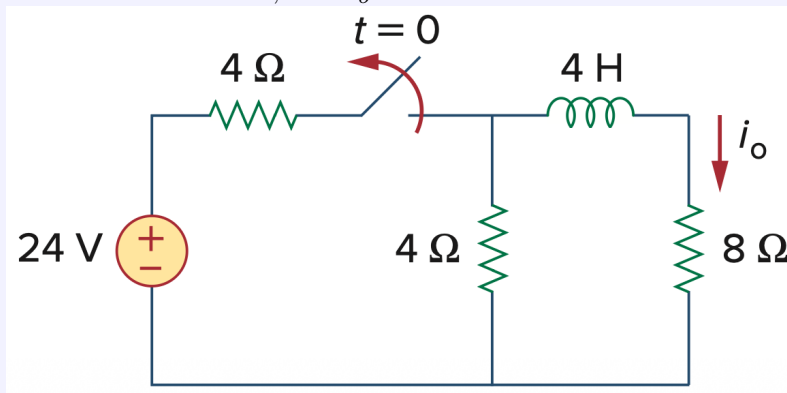
Exercise 9: Time Constant

Determine the time constant for the circuit below.



Exercise 10: Inductor Current Decay

For the circuit below, find i_o for $t > 0$.



Exercise 11: Energy Dissipation during Transient

In the circuit below,

$$v(t) = 80e^{-10^3 t} \text{ V}, \quad t > 0,$$

$$i(t) = 5e^{-10^3 t} \text{ mA}, \quad t > 0.$$

- a) Find R , L , τ .
- b) Calculate the energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.

